

Mapping Africa's Endangered Archaeological Sites and Monuments (MAEASaM): Actor model

Arches Resource Model Graph and Reference Data (Thesaurus)

Version 2

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Quality control
Name

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Schema Attributes and Definitions

All resource model metadata templates in the MAEASaM database, including Remote Sensing Resource Model, are described using the following shared attributes:

Group Name: In Arches, this corresponds to the Semantic Node. It represents the parent category under which related metadata elements are organised.

Node Name: Also referred to as a metadata element. This is the primary descriptive field used to document a Remote Sensing record.

CIDOC CRM Class: Indicates the equivalent class in the CIDOC CRM ontology. This mapping supports semantic interoperability.

CIDOC CRM Property: Indicates the equivalent property in CIDOC CRM ontology. This mapping supports semantic interoperability.

Connector Node: Specifies whether the node establishes a relationship with another resource model. Values are “yes” or “no”.

Connected To: If the Connector Node value is “yes”, this field specifies the name of the related Resource Model.

Data Type: Defines the type of data entered in Arches. Common data types include: string, date, concept list, and resource-instance-list.

Necessity: Indicates whether the node is required when describing a record. Possible values are:

“mandatory”

“strongly suggested”

Cardinality: Specifies how many values a node can contain. In the MAEASaM Arches context, four options are used:

1 - Only one value allowed

0..1 - Optional, maximum one value

1..n - At least one value required, multiple allowed

0..n - Optional, multiple values allowed

Scope Note: Provides a definition and conceptual explanation of the node.

Protocol for Entry: Explains how information should be entered into this node, including formatting rules, conventions, or controlled vocabulary guidance.

Section I: Overview

I.1 Objectives and Scope

The function of the Actor resource model is to record the information related to the different entities that form part of the Mapping Africa's Endangered Archaeological Sites and Monuments project. This model has two primary types of actors, individuals and organisations, and indicates which organisations and individuals are part of the projects and how they relate to one another. Due to the nature of the project and its wide geographic scope, it might also occur that independent researchers might also donate their data to be included in the project. In these cases, the independent researcher will also be recorded within the model.

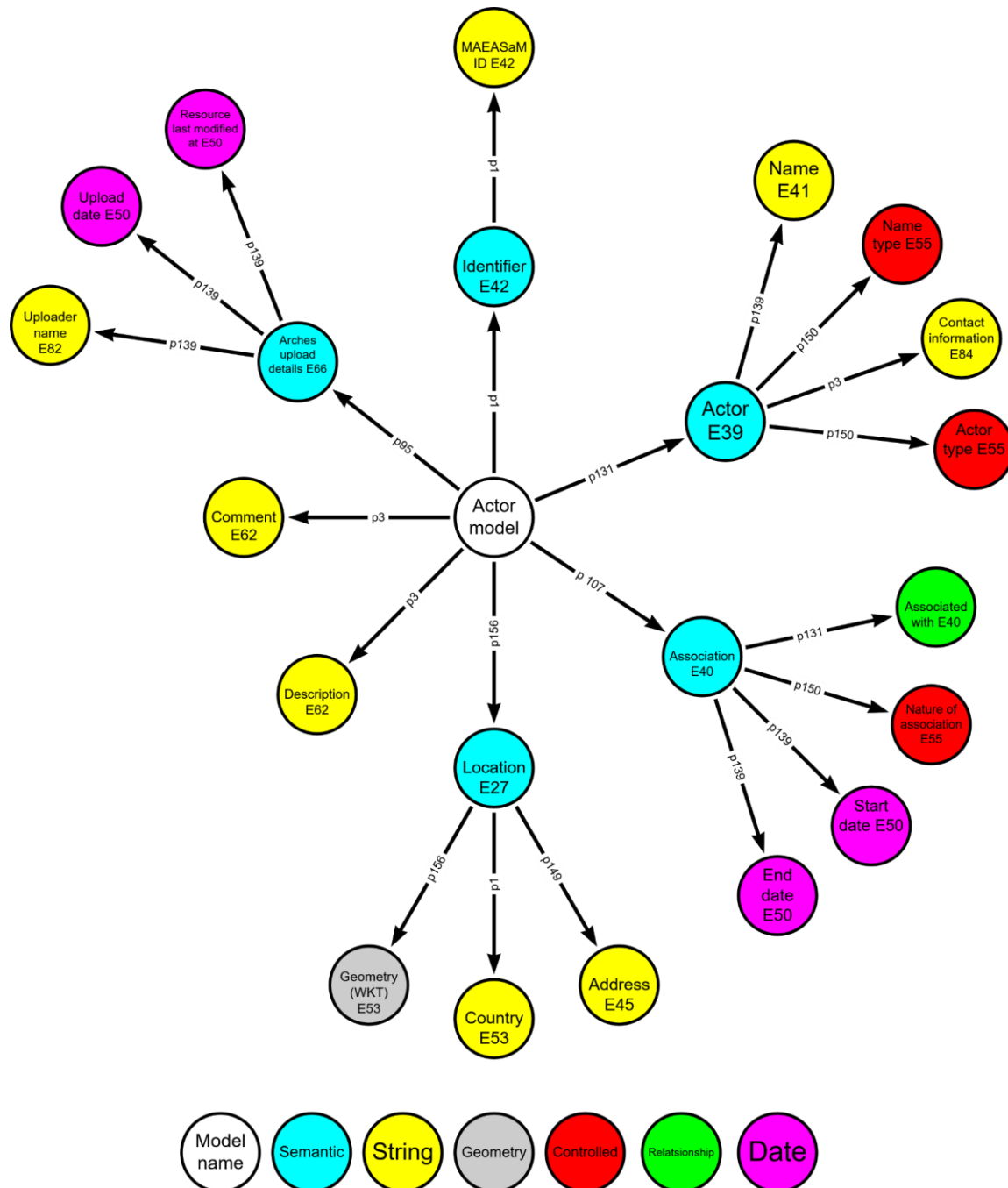
Geographical extent: Dependent on the location of the actor

I.2 History of the creation of the Actor Resource Model

The model was developed to keep track of the different project members, collaborators and independent researchers that contribute to the project. The goal of the model is to keep the author context for each Arches instance within the database. Due to the nature of the project, datasets from various periods and locations can be added, which needs to be linked to the source of the dataset. Therefore, this model records the name of the author, institutional information and the nature of the association between the author of the dataset and the project.

2. Resource model graph

2.1 Structure of the Actor Resource Graph



A graphical representation of the graph structure used in the Actor resource model using the CIDOC-CRM version 6.2 ontology (<https://cidoc-crm.org/Version/version-6.2>). Each colour represents a different data node used within the Arches database, the CIDOC class (E) is used and how that node relates to other nodes via the CIDOC property (p).

2.2 Actor Resource Model Nodes and CIDOC-CRM Structure

Semantic Node: Identifier

Node Name: MAEASaM ID

CIDOC CRM Class: E42 Identifier

CIDOC CRM Property: p1 is identified by

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 1

Necessity: Mandatory

Scope note (description of node)

This node captures the value of the MAEASaM ID for each actor instance and uses the E42 identifier value in the CIDOC-CRM (v6.2) ontology and is linked with the p1 is identified by property. The ID is an alphanumeric identifier separated into two sections, divided by a dash (-). The first section of the ID has the name of the model to which it belongs, and for this model it is represented by "ACT". The numerical part follows sequentially, starting from 1. The design of the ID structure uses a fixed character size, and thus the first number will be preceded by 8 zeros, to create a numerical number consisting of 8 characters. As the sequential number increases, for every 10, 100, 1000 etc, the corresponding zero will be removed to ensure the character length remains 8 characters.

Example:

ACT-00000001

ACT-00000010

ACT-00000100

ACT-00001000

Protocol for entry:

- Allow only one ID per entry.
- Determine the ID value based on the relevant entry.

Semantic Node: Actor

Node Name: Name

CIDOC CRM Class: E41 Appellation

CIDOC CRM Property: p139 has alternative form

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 1..n

Necessity: Mandator

Scope note (description of node)

This node captures the name of the actor that is associated with the Arches instance being added and uses the E41 class with the p139 property. the name value should be added as it will be used within the database. This is a string node which allows the name of the actor to be added in the required structure.

Example

Renier Hendrik van der Merwe

Protocol for entry:

- Determine the structure of the required name. Should match the source (case of legacy data) or how the actor wants the name added

Node Name: Name type

CIDOC CRM Class: E55 type

CIDOC CRM Property: p150 defines typical parts of

Connector Node: No

Connected to: N/A

Data type: Concept-list

Name of concepts collection: Name type

Cardinality: 1..n

Necessity: Mandator

Scope note (description of node)

This node captures the type of name being captured and uses the E55 class with the p150 property. The name type is a controlled vocabulary and the value used must match that present in the vocabulary precisely.

Example

Primary name

Protocol for entry:

- Determine the appropriate value in the vocabulary for the entry being recorded.
- Ensure that the value is the same as the vocabulary, with attention on the following:
 - White space
 - Capitalisation
 - Special characters

Node Name: Contact information

CIDOC CRM Class: E84 Information carrier

CIDOC CRM Property: p3 has note

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 0..n

Necessity: Optional

Scope note (description of node)

This node captures information related to how the individual can be contacted. This can be either email, cellular phone, land telephone etc and uses the E84 class with the p3 property.

Example

john.smith@contact.com

Protocol for entry:

- Add all available contact information to the node.
 - Each type of contact detail should be added as a new card

Node Name: Actor type

CIDOC CRM Class: E55 type

CIDOC CRM Property: p150 defines typical parts of

Connector Node: No

Connected to: N/A

Data type: Concept-list

Name of concepts collection: Actor type

Cardinality: 1..n

Necessity: Mandator

Scope note (description of node)

This node captures the type of name being captured and uses the E55 class with the p150 property. The name type is a controlled vocabulary and the value used must match that present in the vocabulary precisely.

Example

Project team member

Protocol for entry:

- Determine the appropriate value in the vocabulary for the entry being recorded.
- Ensure that the value is the same as the vocabulary, with attention on the following:
 - White space

- Capitalisation
- Special characters

Semantic Node: Association

Node Name: Associated with
CIDOC CRM Class: E40 Legal body
CIDOC CRM Property: p131 is identified by
Connector Node: No
Connected to: N/A
Data type: Resource-instance-List
Name of concepts collection: N/A
Cardinality: 1..n
Necessity: Mandator

Scope note (description of node)

This node captures how the actor is related to another actor and uses the E40 class with the p131 property. An actor can be either an individual or an institution and thus the association can be either between an actor and institution or between multiple institutions.

Example

Arcadia

MAEASaM Project

Protocol for entry:

- The name of the actor a link is being created to needs to be added as it is present in the Arches database.
- Ensure that the value is the same as the vocabulary, with attention on the following:
 - White space
 - Capitalisation
 - Special characters

Node Name: Nature of association
CIDOC CRM Class: E55 type
CIDOC CRM Property: p150 defines typical parts of
Connector Node: No
Connected to: N/A
Data type: Concept-list
Name of concepts collection: Nature of association
Cardinality: 1..n
Necessity: Mandator

Scope note (description of node)

This node captures how the relevant actor relates to the actor in which the relationship link of the previous node is linked to and uses the E55 class with the p150 property.

Example

Project funder

Protocol for entry:

- Determine the appropriate value in the vocabulary for the entry being recorded.
- Ensure that the value is the same as the vocabulary, with attention on the following:
 - White space
 - Capitalisation
 - Special characters

Node Name: Start date

CIDOC CRM Class: E50 Date

CIDOC CRM Property: p139 has alternative form

Connector Node: No

Connected to: N/A

Data type: EDTF

Name of concepts collection: N/A

Cardinality: 1

Necessity: Mandatory

Scope note (description of node)

This node captures the start date on which the actor was part of the project or on which the actor contributed to the project and uses the E50 date with the p139 property.

Example:

2026-02-19

Protocol for entry:

- The date should follow the Gregorian calendar
- Use year, month, and day separated by hyphens (e.g., YYYY-MM-DD)
- Any other format will cause an error during upload

Node Name: End date

CIDOC CRM Class: E50 Date

CIDOC CRM Property: p139 has alternative form

Connector Node: No

Connected to: N/A

Data type: EDTF

Name of concepts collection: N/A

Cardinality: 1

Necessity: Mandatory

Scope note (description of node)

This node captures the end date on which the actor was part of the project or on which the actor contributed to the project and uses the E50 date with the p139 property.

Example:

2026-02-19

Protocol for entry:

- The date should follow the Gregorian calendar
- Use year, month, and day separated by hyphens (e.g., YYYY-MM-DD)
- Any other format will cause an error during upload

Semantic Node: Location

Node Name: Address

CIDOC CRM Class: E45 Address

CIDOC CRM Property: p149 is identified by

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 0..n

Necessity: Optional

Scope note (description of node)

This node captures the physical address of the actor, either the present or when the dataset was created, and uses the E45 class with the p149 property.

Example

University of Pretoria, Lynnwood Rd, Hatfield, Pretoria, 0083

Protocol for entry:

- Address must be added as is provided with the dataset or as requested by the actor
- Unless a specific reason has been given, the institutional address added will be the official address provided by the institution.

Node Name: Country

CIDOC CRM Class: E53 Place

CIDOC CRM Property: p1 is identified by

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 0..n

Necessity: Optional

Scope note (description of node)

This node is an extension of the previous node and captures the location in which the actor is present or was present when the dataset was created and uses the E53 class with the pl property.

Example

Kenya

Protocol for entry:

- The country in which the actor is/was present during the period relevant to the entry should be used

Node Name: Geometry (WKT)

CIDOC CRM Class: E53 Place

CIDOC CRM Property: L54 is same as

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 1..n

Necessity: Mandatory

Scope note (description of node)

This node provides information related to the geographical extent of each entry as it relates to the earth surface. The node uses the E53 class with the L54 (CIDOC CRM 6.2 extension CRMdig v3.2.1) property. The geometry is based on the World Geodetic System (WGS) 84 system, with the longitude and latitude points being presented in Well-known Text (WKT) format. Within the Arches system, geometry is stored as a GeoJSON-feature-collection, with upload formats being Shapefile (SHP), Keyhole Markup Language (KML) and the Geometry JavaScript Object Notation (geoJSON) format. Once uploaded to Arches, all geometry is transformed into geoJSON format.

Example

POLYGON ((-7 25.5,-6.75 25.5,-6.75 25.25,-7 25.25,-7 25.5))

Protocol for entry:

- All geometry values in the row must be in WKT format using the WGS 84.

Node Name: Description

CIDOC CRM Class: E62 String

CIDOC CRM Property: p3 has note

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 1

Necessity: If applicable

Scope note (description of node)

This is a general text node that uses the E62 class with the p3 property. It describes the relevant entry.

Example

Remember to avoid special characters as far as possible.

Protocol for entry:

- This is a general text node
- Special characters should be avoided if possible, including: , " ' % # ; ;

Node Name: Comment

CIDOC CRM Class: E62 String

CIDOC CRM Property: P3 has note

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 0..n

Necessity: If applicable

Scope note (description of node)

This is a general, all-purpose node that captures any information that is regarded as required for the specific entry but is not captured in any of the other nodes present within the resource model. It uses the E62 String with the L54 (CIDOC CRM 6.2 extension CRMdig v3.2.1) property.

Example

Remember to avoid special characters as far as possible.

Protocol for entry:

- This is a general text node
- Special characters should be avoided if possible, including: , " ' % # ;

Semantic Node: Arches upload details

Node Name: Uploader name

CIDOC CRM Class: E82 Actor Appellation

CIDOC CRM Property: p139 has alternative form

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 1

Necessity: Mandatory

Scope note (description of node)

The function of this node is to capture the information related to the name of the individual who uploaded the information of the associated entry to the Arches database, and uses the E82 class with the p139 property.

Example:

Renier Hendrik van der Merwe

Protocol for entry:

- The name of the individual who uploaded the data to the Arches database should be the full version of the name.

Node Name: Upload date

CIDOC CRM Class: E63 Beginning of Existence

CIDOC CRM Property: p92 brought into existence

Connector Node: No

Connected to: N/A

Data type: EDTF

Name of concepts collection: N/A

Cardinality: 1

Necessity: Mandatory

Scope note (description of node)

The function of this node is to capture the date on which the information of the associated entry was uploaded to the Arches database and uses the E62 class with the p92 property.

Example:

2026-02-19

Protocol for entry:

- The date should follow the Gregorian calendar
- Use year, month, and day separated by hyphens (e.g., YYYY-MM-DD)
- Any other format will cause an error during upload

Node Name: Resource last modified at
CIDOC CRM Class: E63 Beginning of Existence
CIDOC CRM Property: p92 brought into existence
Connector Node: No
Connected to: N/A
Data type: EDTF
Name of concepts collection: N/A
Cardinality: 1
Necessity: Mandatory

Scope note (description of node)

The function of this node is to capture the date on which any modification was made to any nodes related to the associated entry. This node uses the E63 class with the p92 property.

Example:
2026-03-05

Protocol for entry:

- Every time any modification is made to a specific entry within the resource model, the date should be captured as a new value
- All dates already present should be kept and unmodified
- The date should follow the Gregorian calendar
- Use year, month, and day separated by hyphens (e.g., YYYY-MM-DD)
- Any other format will cause an error during upload

Section 3: Comments

References

Aalber, T., Balze, D., Bekiari, C., Boudouri, L., Crofts, N., Dunsire, G., Eide, Ø., Gill, T., Goerz, G., Hagedorn-Saupe, M., Hiebel, G., Holmen, J., Inkari, J., Iorizzo, D., Kotipelt, J., Kraus, S., Lampe, K.H., Lamsfus, C., Lindenthal, J., Nyman, M., Riva, P., Rold, L., Smiraglia, R., Stein, R., Stiff, M and Žumer, M. 2015. *Definition of the CIDOC Conceptual Reference Mode*. https://cidoc-crm.org/sites/default/files/cidoc_crm_version_6.2.pdf. Access: 2026-03-12